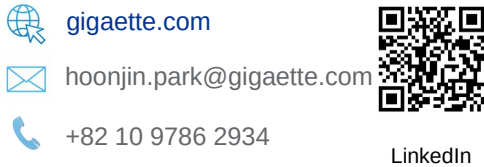


Contact Information



LinkedIn

Team

Daejun CHANG | Co-Founder & CEO

- Professor of Mechanical Engineering, KAIST
- Ph.D. Chemical Engineering, KAIST; ex-Hyundai Heavy Industries
- Founder, Lattice Technology
- Leads national R&D programs and overseas government relations

Sungkwang LEE | CTO

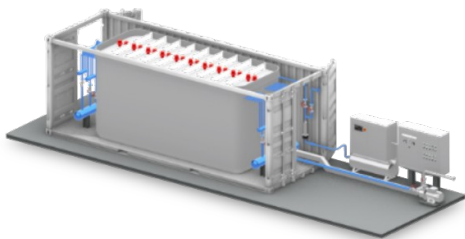
- Leads system design, operation and application development
- M.S. Mechanical & Aerospace Eng., Seoul National University; KAIST Finance MBA
- Former engineer, LG Electronics

Hoonjin PARK | Co-Founder & COO

- Leads sales, partnerships and global business development
- Co-founder of Lattice Technology and Recreate
- KAIST Finance MBA

Jinkwang LEE | Technical Advisor

- Ph.D. Mechanical Engineering, KAIST; Assoc. Prof., Gyeongsang National Univ.
- Ex-DSME, Korean Register and Hyundai Heavy Industries



IsoTES-TEU 2 containerized thermal battery



12 MWh IsoTES for district heating, Mongolia

Problem

- **Waste heat is dumped:** exhaust, cooling-water and incineration heat is lost because its timing and temperature rarely match demand
- **Processes are rigid:** batch heat cannot feed continuous loads; boilers are oversized for peaks, with no thermal backup
- **Fossil boilers lock in emissions:** LNG and coal heat drives Scope 1 under CBAM/RE100 pressure; hydrogen heat is costly and risky
- **Energy bills stay high:** peak power and demand charges inflate costs; Li-ion ESS is too costly and short-lived for bulk storage

Solution

- IsoTES®: single-tank liquid thermal battery charged by waste heat or surplus / off-peak electricity
- Discharges steam, hot air, heating or cooling at the exact temperature and pressure a process needs
- Standard modules (2-TEU container, 40–500 m³ silos, 200 m³ box) scale from 0.1 MWh to GWh-class plants

Key Features

- Patented "thermal piston": ~90% of discharge at constant temperature vs. ~20% for brick-based rivals
- High-pressure output (up to 250 bar) via in-tank heat-transfer tubes; no extra blowers or compressors
- ~1/3 the CAPEX and ~4x the lifetime of Li-ion ESS (30 vs. 8 years), 95% power-to-heat efficiency
- Wide range: -80 °C to +550 °C today, roadmap to 1,000 °C and 1 GWh

Business Model

- Now: sales of IsoTES modules and integrated packages (TES + waste-heat recovery, heat pump, chiller) with O&M
- From 2028: Heat-as-a-Service / Storage-as-a-Service; GIGAette funds the asset, customers pay per unit of heat
- From 2030: industrial-park and city-scale heat platforms (B2B/B2G) and overseas licensing

Business Snapshot

- Raised KRW 1.8B (~USD 1.3M) equity: KRW 0.9B Seed (Sep 2025) from Shinhan Venture Investment, Bluepoint Partners, Daejeon CCEI and Star Ventures; KRW 0.9B from Sunbo Angel Partners (Apr 2026)
- Secured KRW 3.5B (~USD 2.5M) non-dilutive R&D funding: Deep Tech TIPS (KRW 1.5B, 2025) and Scale-up TIPS (KRW 2.0B, 2026)
- First revenue: KRW 173M 3-TPD CO₂ liquefaction system for the Institute for Advanced Engineering (Aug 2026)
- Track record: 0.1 MWh pilot (2023), 0.7 MWh prototype (2024), 1.5 MWh Mongolia PoC (2024), 12 MWh KOICA project (2026)
- Pipeline: YK Steel (EAF waste heat), ALUS and KONEC (aluminum waste heat to cooling), Mongolian district heating operators
- 8 KR/PCT patents incl. exclusive KAIST license; 5 MWh multi-PoC facility (MSS-funded, 2025–2028)

Seeking Opportunities

- **Renewable-powered cooling:** buildings, data centers and factories in hot climates cutting power bills via stored renewable heat + absorption chillers
- **Renewable-powered heat:** grain drying and agri / industrial process heat users in Southeast Asia
- **Local partners:** EPCs, distributors and developers for IsoTES pilots
- **Investors:** strategic and climate-tech investors for Series A (2027)